

Office hours

11/25

SVD: $X = UDV^T$
 $n \times p \quad n \times n \quad p \times p \quad p \times p$

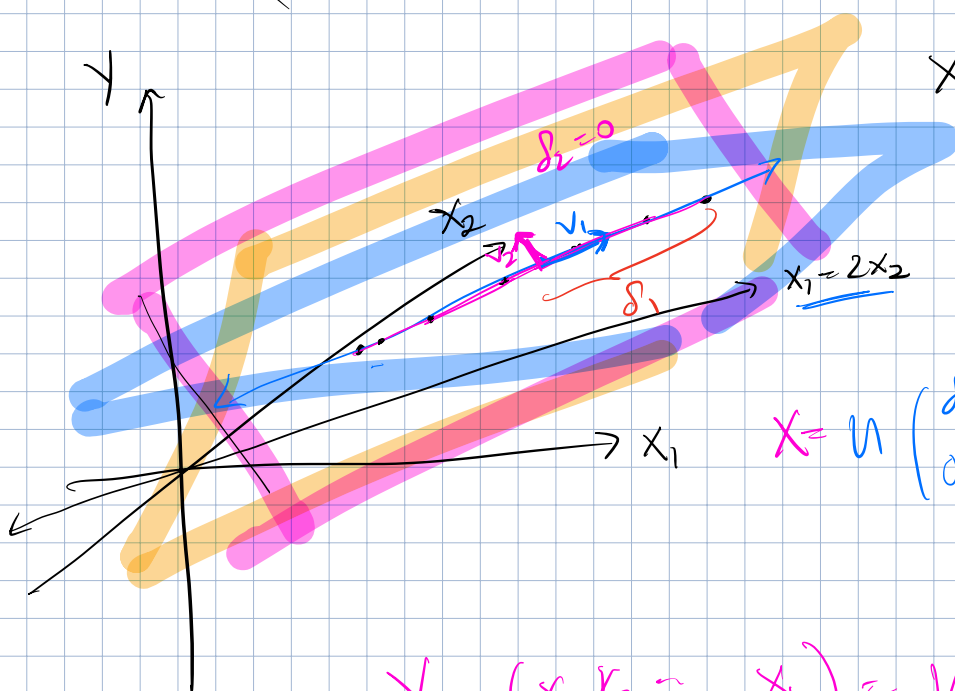
U is orthonormal

D is diag

V is orthonormal

$$D = \begin{pmatrix} \delta_1 & & \\ & \delta_2 & \\ & & \ddots \\ & & & \delta_p \end{pmatrix}$$

sd of the spread of the data in the v_j th direction



$$X = \begin{pmatrix} 1 & 1 \\ x_1 & x_2 \end{pmatrix}$$

$x_1 = 2x_2$

$$X = U \begin{pmatrix} \delta_1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix}$$

$$X = (x_1, x_2, \dots, x_p) = U \begin{pmatrix} \delta_1 & \delta_2 & \dots \\ & & \ddots \\ & & & \delta_p \\ & & & & \approx 0 \end{pmatrix}$$

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T Y = (U D V^T)^T (U D V^T)^{-1} (U D V^T)^T Y \\ &= (V D U^T U D V^T)^{-1} (V D U^T Y) \\ &= (V D^2 V^T)^{-1} V D U^T Y \end{aligned}$$

$$= V D^{-2} \cancel{V^{-1} V} D U^T Y$$

$$= V D^{-2} D U^T Y = V \overset{\text{non-invertible if not all } d_j \neq 0}{D^{-1}} U^T Y$$

$$\text{Var}(\hat{\beta}) = (X^T X)^{-1} \sigma^2$$

$$= V D^{-2} V^T \sigma^2$$

↑ blows up as $\min_j d_j \rightarrow 0$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\hat{y} = X \hat{\beta} = H Y$$

$$e = Y - \hat{y} = (I - H) Y$$

$$Y \sim N(X\beta, \sigma^2 I) ?$$

$$Y = X\beta + \varepsilon \quad \text{we know } \varepsilon \sim N(0, \sigma^2 I)$$

$$\begin{aligned} E(Y) &= E(X\beta) + E(\varepsilon) \\ &= X\beta + 0 = \underline{X\beta} \end{aligned}$$

$$\text{Var}(Y) = \text{Var}(X\beta + \varepsilon) = \text{Var}(\varepsilon) = \underline{\sigma^2 I}$$

Show $\min_{\beta} Q(\beta) = Y^T (I - H) Y$.

Def:

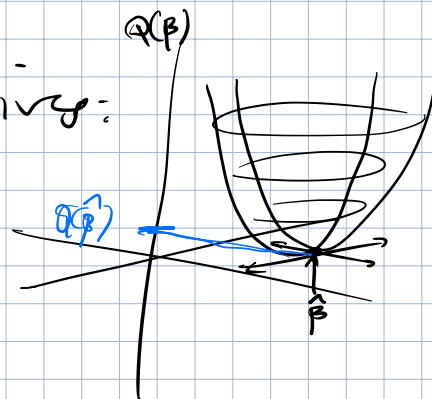
$$Q(\beta) = \|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta)$$

Solving the LS minimization problem gives:

$$\frac{\partial Q}{\partial \beta} \stackrel{!}{=} 0$$

$\Leftrightarrow$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$



so the min value of Q is $Q(\hat{\beta})$

$$Q(\hat{\beta}) = \|Y - X\hat{\beta}\|^2 = \|Y - X(X^T X)^{-1} X^T Y\|^2$$

$$= \|Y - HY\|^2$$

$$= \|(I - H)Y\|^2$$

$$= [(I - H)Y]^T [(I - H)Y]$$

$$= Y^T (I - H)^T (I - H) Y$$

$$= Y^T (I - H) Y$$

$\square$

claim: $(I - H)^T = (I - H)$
 $(I - H)^2 = (I - H)$

$$\begin{aligned} \textcircled{1} \quad (I - H)^T &= I^T - H^T = I - (X(X^T X)^{-1} X^T)^T \\ &= I - (X^T)^T [(X^T X)^{-1}]^T X^T \\ &= I - X [(X^T X)^{-1}]^T X^T \\ &= I - X (X^T X)^{-1} X^T = I - H \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad & \underbrace{(I-H)}^1 \underbrace{(I-H)}^1 = I \cdot I - I \cdot H - H \cdot I + H \cdot H \\
 & = I - H - H + H^2 \\
 & = I - 2H + \underbrace{X(X^T X)^{-1} X^T}_{\cancel{X(X^T X)^{-1} X^T}} \\
 & = I - 2H + H = I - H \checkmark
 \end{aligned}$$

Claim:

$$F = \frac{(\hat{\beta} - \beta)^T (X^T X) (\hat{\beta} - \beta) / p}{Y^T (I - H) Y / (n - p)} \sim F_{p, n-p}$$

$$\begin{aligned}
 \textcircled{1} \quad & \text{Num} : \sim \chi_p^2 / p \\
 \textcircled{2} \quad & \text{Den} : \sim \chi_{n-p}^2 / (n-p)
 \end{aligned}
 \left. \vphantom{\begin{aligned} \textcircled{1} \\ \textcircled{2} \end{aligned}} \right\} \text{meina}$$

③ Num & Den $\leftarrow$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\hat{\beta} - \beta \sim N(0, \sigma^2 (X^T X)^{-1})$$

$$\underbrace{[\sigma^2 (X^T X)^{-1}]^{-1/2}}_{\parallel} (\hat{\beta} - \beta) \sim N(0, [\sigma^2 (X^T X)^{-1}]^{-1/2} \sigma^2 (X^T X)^{-1} (\sigma^2 (X^T X)^{-1})^{1/2})$$

$$(X^T X)^{1/2} (\hat{\beta} - \beta) / \sigma \sim N(0, I_p)$$

$\uparrow$

$$\underline{z} \sim N(0, \underline{V})$$

$$\underline{V}^{-1/2} \underline{z} \sim N(0, \underline{I})$$

$$E(\underline{V}^{-1/2} \underline{z}) = \underline{V}^{-1/2} E(\underline{z})$$

$$\begin{aligned} \text{Var}(\underline{V}^{-1/2} \underline{z}) &= \underline{V}^{-1/2} \text{Var}(\underline{z}) (\underline{V}^{-1/2})^T \\ &= \underline{V}^{-1/2} \underline{V} \underline{V}^{-1/2} \\ &= \cancel{\underline{V}^{-1/2}} \cancel{\underline{V}^{1/2}} \cancel{\underline{V}^{1/2}} \cancel{\underline{V}^{-1/2}} \\ &= \underline{I} \end{aligned}$$

$$\begin{aligned} \|(X^T X)^{1/2} (\hat{\beta} - \beta) / \sigma\|^2 &= (\hat{\beta} - \beta)^T (X^T X)^{1/2} (X^T X)^{1/2} (\hat{\beta} - \beta) / \sigma^2 \\ &= (\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta) / \sigma^2 \sim \chi_p^2 \end{aligned}$$

$$\text{If } \underline{z} \sim N(0, \underline{I}_k) \Rightarrow \|\underline{z}\|^2 \sim \chi_k^2$$

$$\text{Num: } (\hat{\beta} - \beta)^T (X^T X) (\hat{\beta} - \beta) / p \sigma^2 \sim \chi_p^2 / p$$

$$\begin{aligned} \text{Den: } Y^T (I - H) Y &= Y^T (I - H)^T (I - H) Y \\ &= \|(I - H) Y\|^2 \\ &= \|\underline{e}\|^2 = \underline{e}^T \underline{e} = \text{SSE} \end{aligned}$$

$$\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2$$

$$\frac{Y^T(I-H)Y}{\sigma^2} \sim \chi^2_{n-p}$$

$$\frac{\hat{\sigma}^2}{\sigma^2} = \frac{Y^T(I-H)Y}{\sigma^2(n-p)} \sim \chi^2_{n-p}/(n-p)$$

$$F = \dots = \frac{x_p^2/p}{x_{n-p}^2/(n-p)} \stackrel{\text{if } N \perp D \checkmark}{\sim} F_{p, n-p}$$

Idea: If $\hat{\beta} \perp e$ *

then

$\underline{g(\hat{\beta})} \perp \underline{h(e)}$ for any deterministic $g(\cdot)$ & $h(\cdot)$

so

$$(\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta) / \sigma^2 \perp \|e\|^2 = e^T e = \text{Denom.}$$

Claim:
Why is

Num

$$\hat{\beta} \perp e?$$

$$Y \sim N(X\beta, \sigma^2 I)$$

$$\begin{pmatrix} \hat{\beta} \\ e \end{pmatrix} = \underbrace{\begin{pmatrix} (X^T X)^{-1} X^T \\ I - H \end{pmatrix}}_{A} Y$$

$$\begin{pmatrix} \hat{\beta} \\ e \end{pmatrix} = AY \sim N(AX\beta, A\sigma^2 I A^T)$$

$$\sim N \left(\begin{pmatrix} (X^T X)^{-1} X^T \\ I - H \end{pmatrix} \beta, \sigma^2 \begin{pmatrix} (X^T X)^{-1} X^T \\ \underline{I - H} \end{pmatrix} \begin{pmatrix} \underline{(X^T X)^{-1} X^T}^T \\ (I - H)^T \end{pmatrix} \right)$$

$$\sim N \left(\begin{pmatrix} (X^T X)^{-1} X^T \beta \\ (I - H) X \beta \end{pmatrix}, \right)$$

$$\sim N \left(\begin{pmatrix} \beta \\ 0 \end{pmatrix}, \sigma^2 \begin{bmatrix} \cancel{(X^T X)^{-1} X^T X (X^T X)^{-1}} & \cancel{(X^T X)^{-1} X^T (I - H)^T} \\ \underline{(I - H) X (X^T X)^{-1}} & \underline{(I - H) (I - H)^T} \end{bmatrix} \right)$$

$$\sim N \left(\begin{pmatrix} \beta \\ 0 \end{pmatrix}, \sigma^2 \begin{pmatrix} (X^T X)^{-1} & 0 \\ \underline{(I - H) X (X^T X)^{-1}} & \underline{(I - H)} \end{pmatrix} \right)$$

$$HX = X \cancel{(X^T X)^{-1} X^T} X = X$$

$$\begin{pmatrix} \hat{\beta} \\ e \end{pmatrix} \sim N \left(\begin{pmatrix} \beta \\ 0 \end{pmatrix}, \sigma^2 \begin{pmatrix} (X^T X)^{-1} & \underline{0} \\ \underline{0} & I - H \end{pmatrix} \right)$$

$\hat{\beta}$ & e are jointly normal w/

$\text{cov}(\hat{\beta}, e) = 0 \Rightarrow \hat{\beta} \perp e$

*claim

$$\begin{aligned}
 \text{Var}(\underline{e}) &= \text{Var}((I-H)Y) = (I-H)\text{Var}(Y)(I-H)^T \\
 &= \sigma^2(I-H)I(I-H) \\
 &= \sigma^2(I-H)
 \end{aligned}$$

$$\boxed{\text{Var}(\underline{e}_i)} = \sigma^2(1 - \underline{h_{ii}})$$

$$= \sigma^2(1 - x_i(X^TX)^{-1}x_i^T)$$

$$\begin{aligned}
 \text{SLR} \\
 = \underline{\sigma^2} \left(1 - \frac{1}{n} - \frac{\overset{\downarrow}{(x_i - \bar{x})^2}}{SSX} \right)
 \end{aligned}$$